

12 February
Thursday

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Types of Random process

A random process $\{x(t)\}$ is a wide sense stationary if the following conditions are satisfied.

- * $E[x(t)]$ is a constant
- * $E[x(t) \cdot x(t+\tau)]$ a function of τ ($t_1 - t_2$)
(+ difference)

A random process $x(t)$ is defined by $x(t) = \cos \omega t$, where ω is a ~~ran~~ uniform random variable in $(0, \omega_0)$.
Check whether the process is stationary.

$$f_{\omega}(\omega) = \frac{1}{\omega_0 - 0} = \frac{1}{\omega_0}; 0 \leq \omega \leq \omega_0$$

$$E[x(t)] = E(\cos \omega t)$$

$$= \int_0^{\omega_0} \cos \omega t \, d\omega \times f_{\omega}(\omega)$$

$$= \left[\frac{\sin \omega t}{\omega t} \right]_0^{\omega_0} \times f_{\omega}(\omega)$$

$$= \frac{\sin \omega_0 t}{\omega_0 t} - \frac{\sin 0}{t}$$

$$= \frac{\sin \omega_0 t}{\omega_0 t}$$

Not a constant but function of t
 $\therefore x(t)$ is not stationary

A random process is defined by $x(t) = \cos(t + \phi)$ where ϕ is a random variable with pdf equal to $1/\pi$, $-\pi/2 < \phi < \pi/2$. Check whether the process is stationary or not.

$$E[x(t)] = E[\cos(t + \phi)] = \int_{-\pi/2}^{\pi/2} \cos(t + \phi) \times \frac{1}{\pi} d\phi$$

$$= \left[\sin(t + \phi) \right]_{-\pi/2}^{\pi/2}$$

$$= \frac{1}{\pi} \left[\sin(t + \pi/2) - \sin(t - \pi/2) \right]$$

$$= \frac{1}{\pi} \left[\cos(t) + \cos(t) \right]$$

$$= \frac{2\cos(t)}{\pi}$$

not a constant, but a function of t

$\therefore x(t)$ is not stationary

A random process is given by $x(t) = A \sin \omega t$, where ω is a constant and A is random variable uniformly distributed over $(-1, 1)$. Prove that the process is not stationary.

$$f_A(a) = \frac{1}{1 - (-1)} = \frac{1}{2}, \quad -1 < a < 1$$

$$E[x(t)] = E[A \sin \omega t] = (t) x =$$

$$[(0) \sin \omega t \times E(A)] =$$

$$= \sin \omega t \times \int_{-1}^1 a \cdot \frac{1}{2} da$$

$$= \sin \omega t \times \left[\frac{a^2}{2} \right]_{-1}^1 =$$

$$[(0) \sin \omega t] = \sin \omega t \times \left[\frac{1}{2} - \frac{1}{2} \right] =$$

$$[(0) \sin \omega t] = 0, \text{ a constant.}$$

$$R_x(\tau) = E[x(t) \cdot x(t+\tau)]$$

$$= E[A \sin \omega t \cdot A \sin \omega(t+\tau)]$$

$$[(0) + (0+\tau)] = \sin \omega t \cdot \sin \omega(t+\tau) \cdot E[A^2]$$

$$= \sin \omega t \cdot \sin \omega(t+\tau) \cdot \int_{-1}^1 A^2 \cdot \frac{1}{2} dA$$

$$[(0) + (0+\tau)] = \sin \omega t \cdot \sin \omega(t+\tau) \cdot \left[\frac{A^3}{3} \right]_{-1}^1$$

$$= \sin \omega t \cdot \sin \omega(t+\tau) \times \left[\frac{A^3}{3} \right]_{-1}^1$$

$$= \frac{1}{3} \sin \omega t \cdot \sin \omega(t+\tau) \quad [\text{not should be there}]$$

$\therefore x(t)$ is not stationary

Show that the random process given by $X(t) = A \cos(\omega_0 t + \theta)$ where A & θ are constants and θ is uniformly distributed over $(0, 2\pi)$ is a wide sense stationary.

$$f_{\theta}(\theta) = \frac{1}{2\pi - 0} = \frac{1}{2\pi} \quad (0 \leq \theta \leq 2\pi)$$

$$E[X(t)] = E[A \cos(\omega_0 t + \theta)]$$

$$= E[A] \cdot E[\cos(\omega_0 t + \theta)]$$

$$= \int_0^{2\pi} A \cdot \frac{1}{2\pi} \cos(\omega_0 t + \theta) d\theta$$

$$= \frac{A}{2\pi} \int_0^{2\pi} \cos(\omega_0 t + \theta) d\theta$$

$$= \frac{A}{2\pi} \left[\sin(\omega_0 t + 2\pi) - \sin(\omega_0 t) \right]$$

$$= \frac{A}{2\pi} \left[\sin(\omega_0 t) - \sin(\omega_0 t) \right]$$

$$= 0$$

$$R_X(\tau) = E[A \cos(\omega_0 t + \theta) \cdot A \cos(\omega_0(t + \tau) + \theta)]$$

$$= \frac{A^2}{2\pi} \int_0^{2\pi} \cos(\omega_0 t + \theta) \cos(\omega_0(t + \tau) + \theta) d\theta$$

find how to do

evolutionary process

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Evolutionary process

A process that is not stationary in any sense is called evolutionary process. In other words, we can say that a random process is evolutionary if the mean and variance are function of t . [Dependant on t]

moving average

$$X(t) = \frac{1}{T} \int_{t-T}^t X(\tau) d\tau$$

$$E[X(t)] = \frac{1}{T} \int_{t-T}^t E[X(\tau)] d\tau$$

$$X(t) = \frac{1}{T} \int_{t-T}^t A \cos(\omega\tau + \theta) d\tau$$

$$E[X(t)] = \frac{1}{T} \int_{t-T}^t E[A \cos(\omega\tau + \theta)] d\tau$$

$$E[X(t)] = \frac{1}{T} \int_{t-T}^t E[A] \cos(\omega\tau + \theta) d\tau$$

$$E[X(t)] = \frac{1}{T} \int_{t-T}^t A \cos(\omega\tau + \theta) d\tau$$

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Ergodicity

Time average

$$\bar{x}_T = \int_{-T}^T x(t) \times \frac{1}{2T} dt$$

$$= \frac{1}{2T} \int_{-T}^T x(t) dt$$

Note:

A random process is ergodic if time average $\bar{x}$ of the random sample $x(t)$ tends to average of the ensemble μ of the process as $T \rightarrow \infty$.

$$\frac{1}{2T} \int_{-T}^T x(t) dt \rightarrow \mu \text{ as } T \rightarrow \infty$$

where $\mu = \text{ensemble average} = E[x(t)]$

A random process is given by $x(t)$; $x(t) = A \cos(\omega t + \theta)$ where θ is a random variable uniformly distributed over $(0, 2\pi)$. The process is m.s.

* Ensemble
Ergodicity

$$\begin{aligned}
 &= \frac{A}{2\pi} \left[\sin(\omega_0 t + \theta) \right]_0^{2\pi} \\
 &= \frac{A}{2\pi} \left[\sin(\omega_0 t + 2\pi) - \sin(\omega_0 t + \theta) \right] \\
 &= \frac{A}{2\pi} \left[\sin(\omega_0 t) - \sin(\omega_0 t) \right] \\
 &= 0
 \end{aligned}$$

* Time average

$$\bar{x}_t = \frac{1}{2T} \int_{-T}^T x(t) dt$$

$$= \frac{1}{2T} \int_{-T}^T A \cos(\omega_0 t + \theta) dt$$

$$= \frac{A}{2T} \int_{-T}^T \frac{\cos(\omega_0 t + \theta)}{\omega_0} \Big|_{-T}^T + \int_{-T}^T \sin$$

$$= \frac{A}{2T} \left[\frac{\sin(\omega_0 T + \theta) - \sin(-\omega_0 T + \theta)}{\omega_0} \right]$$

$$= \frac{A}{2T\omega_0} \left[\sin(\omega_0 T + \theta) + \sin(\omega_0 T + \theta) \right]$$

Since time average is equal to ensemble average (0), $x(t)$ is mean ergodic

A random process is given
 $X(t) = \cos(t + \phi)$ where ϕ is uniformly distributed over $(0, 2\pi)$. Show that the process is stationary in WSS. Show that the process is mean-ergodic.

$$f_X(t) = \frac{1}{2\pi - 0} = \frac{1}{2\pi}$$

$$\begin{aligned} E(X(t)) &= E[\cos(t + \phi)] \\ &= \frac{1}{2\pi} \int_0^{2\pi} \cos(t + \phi) d\phi \\ &= \frac{1}{2\pi} \left[\sin(t + \phi) \right]_0^{2\pi} \\ &= \frac{1}{2\pi} \left[\sin(2\pi + t) - \sin(2\pi t) \right] \\ &= \frac{1}{2\pi} [0] \end{aligned}$$

$$\begin{aligned} R_X(t) &= E[\cos(t + \phi) \cos((t + \tau) + \phi)] \\ &= \frac{1}{2} \left[\cos(2t + \tau + 2\phi) + \cos(-\tau) \right] \\ &= \frac{\cos(\tau)}{2} + \frac{1}{2} \int_0^{2\pi} \cos(2t + \tau + 2\phi) \times \frac{1}{2\pi} d\phi \\ &= \frac{\cos(\tau)}{2} + \frac{1}{4\pi} \left[\sin(2t + \tau + 2\phi) \right]_0^{2\pi} \\ &= \frac{\cos(\tau)}{2} + \frac{1}{4\pi} \left[\sin(2t + \tau + 4\pi) - \sin(2t + \tau) \right] \end{aligned}$$

$$= \frac{\cos(z)}{2} + \frac{1}{4\pi} \Big|_0^z$$

$$\Rightarrow \frac{\cos(z)}{2}$$

Therefore, $x(t)$ is a WSS.

$$\bar{X}(t) = \frac{1}{2T} \int_{-T}^T \cos(t+\phi) d\phi$$

$$= \frac{1}{2T} \left[\sin(t+\phi) \right]_{-T}^T$$

$$= \frac{1}{2T} \left[\sin(T+\phi) - \sin(-T+\phi) \right]$$

$$= \frac{1}{2T} \left[2 \cos\left(\frac{T+\phi-T+\phi}{2}\right) \sin\left(\frac{T+\phi+T-\phi}{2}\right) \right]$$

$$= \frac{1}{T} \left[\cos\left(\frac{2\phi}{2}\right) \sin\left(\frac{2T}{2}\right) \right]$$

$$= \frac{1}{T} \left[\cos(\phi) \sin(T) \right]$$

Applying $t \rightarrow \infty$,

$$\frac{1}{\infty} \left[\cos(\phi) \sin(\infty) \right]$$

$$= 0$$

Since time average = ensemble average
 $x(t)$ is mean ergodic.

A random process is given by
 $X(t) = A \cos(\omega t) + B \sin(\omega t)$ where A &
 B are random variables such
 that $E[A] = 0$, $E[B] = 0$, $E[AB] = 0$
 $E[A^2] = E[B^2]$. Prove that the process
 is mean ergodic.

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T X(t) dt = \langle X \rangle$$

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T (A \cos(\omega t) + B \sin(\omega t)) dt =$$

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left[\frac{A}{\omega} \sin(\omega t) - \frac{B}{\omega} \cos(\omega t) \right]_0^T =$$

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left[\frac{A}{\omega} \sin(\omega T) - \frac{B}{\omega} \cos(\omega T) - \frac{A}{\omega} \sin(0) + \frac{B}{\omega} \cos(0) \right] =$$

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left[\frac{A}{\omega} \sin(\omega T) - \frac{B}{\omega} \cos(\omega T) + \frac{B}{\omega} \right] =$$

$$\lim_{T \rightarrow \infty} \left[\frac{A}{\omega T} \sin(\omega T) - \frac{B}{\omega T} \cos(\omega T) + \frac{B}{\omega T} \right] =$$

$$\lim_{T \rightarrow \infty} \left[\frac{A}{\omega T} \sin(\omega T) - \frac{B}{\omega T} \cos(\omega T) + \frac{B}{\omega T} \right] =$$

$$\lim_{T \rightarrow \infty} \left[\frac{A}{\omega T} \sin(\omega T) - \frac{B}{\omega T} \cos(\omega T) + \frac{B}{\omega T} \right] =$$

$$0 =$$

Since time average = ensemble average
 the process is mean ergodic.

Markov Chain

A random process in which the set of states, space 'S' and the set of parameter 'T' are discrete taking only a finite or countable number of finite values in which the probability P_{ij} , that the process would go from i^{th} state to j^{th} state in the next step depends only on the present state and not on how it arrived at that state is called Markov Chain of the first order.

$$X_{n-1} = i_{n-1}$$

$$P[X_{n+1} = j \mid X_0 = i_0, X_1 = i_1, X_2 = i_2, \dots, X_n = i_n]$$

$$P[X_{n+1} = j \mid X_n = i] \quad P[\text{today} \mid \text{past values}]$$

Let $X_0, X_1, \dots, X_{n-1}$ are past values

X_n is present value

X_{n+1} is future value

One step transition probability matrix

	States of X_{n+1}					
	0	1	2	3	...	n
States of X_n	0	P_{00}	P_{01}	P_{02}	P_{03}	= 1
	1	P_{10}	P_{11}	P_{12}	P_{13}	= 1
	2	P_{20}	P_{21}	P_{22}	P_{23}	= 1
	:					
	n				P_{nn}	

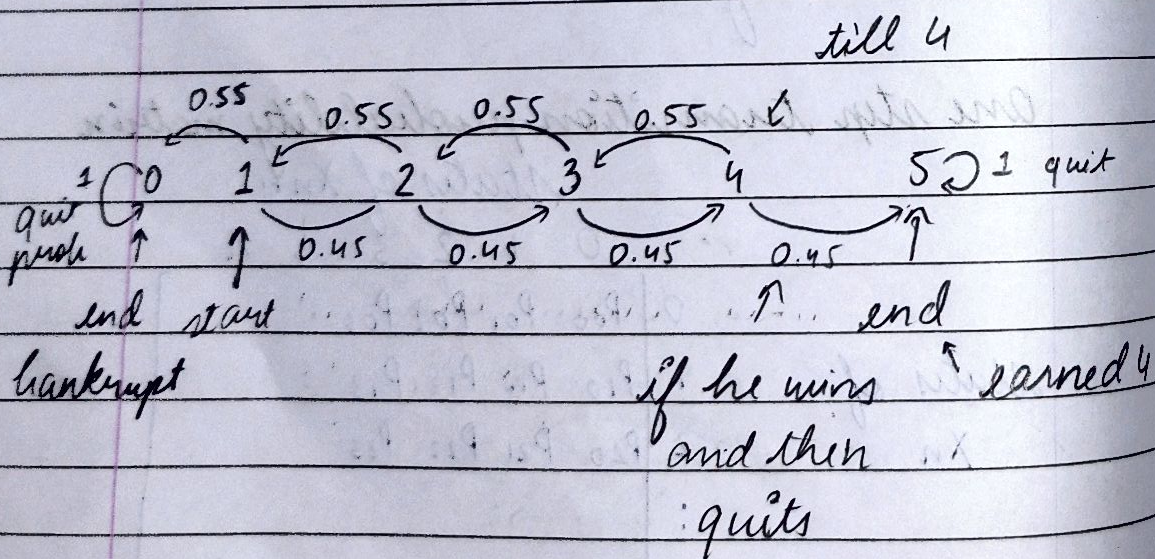
n step TPM

Rules of one step transition prob. matrix

1. $p_{ij} \geq 0$; $i, j \geq 0$ prob never neg
2. $\sum_{i=0}^{\infty} p_{ij} = 1$; $i=0$, prob sum = 1

Suppose in a gambling game a gambler wins rupees 1 with probability 0.45 and loses rupees 1 with probability 0.55 at any state. He starts the game with rupees 1 and quits when either he has earned rupees 4 or he is bankrupt write the transition probability matrix.

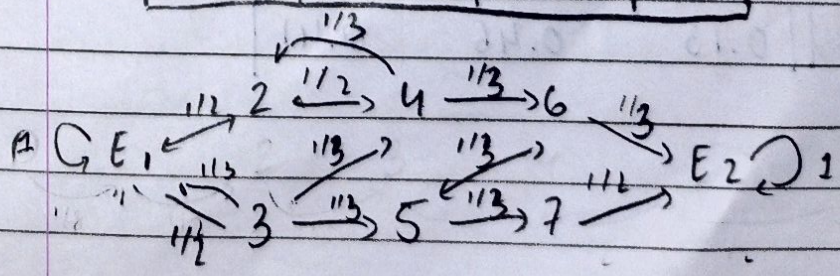
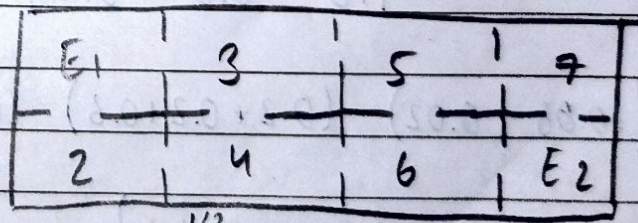
Event diagram



Transition probability matrix X_{n+1}

	0	1	2	3	4	5
0	1	0	0	0	0	0
1	0.55	0	0.45	0	0	0
2	0	0.55	0	0.45	0	0
3	0	0	0.55	0	0.45	0
4	0	0	0	0.55	0	0.45
5	0	0	0	0	0	1

Consider an experiment in which a rat is wandering inside a maze which has eight compartments $\{E_1, E_2, E_3, E_4, E_5, E_6, E_7, E_8\}$. Each compartment has doors as shown in figure. If the rat reaches the compartment E_1, E_2 it escapes otherwise it moves from one compartment to the other through the doors at random. Write the transition probability matrix and transition figure.



	t_1	2	3	4	5	6	7	t_2
t_1	1	0	0	0	0	0	0	0
2	$\frac{1}{2}$	0	0	$\frac{1}{2}$	0	0	0	0
3	$\frac{1}{3}$	0	0	$\frac{1}{3}$	$\frac{1}{3}$	0	0	0
4	0	$\frac{1}{3}$	$\frac{1}{3}$	0	0	$\frac{1}{3}$	0	0
5	0	0	$\frac{1}{3}$	0	0	$\frac{1}{3}$	$\frac{1}{3}$	0
6	0	0	0	$\frac{1}{3}$	$\frac{1}{3}$	0	0	$\frac{1}{3}$
7	0	0	0	0	$\frac{1}{2}$	0	0	$\frac{1}{2}$
t_2	0	0	0	0	0	0	0	1

Initial conditions

$$\begin{bmatrix} 0.4 & 0.4 & 0.2 \\ 0.8 & 0.4 & 0.5 \\ 0.0 & 0.6 & 0.4 \end{bmatrix}$$

$$\mu(1) = 0.2$$

$$\mu(2) = 0.5$$

$$\mu(3) = 0.3$$

Find the probabilities of the process at $T=1$

$$\begin{bmatrix} 0.2 & 0.5 & 0.3 \end{bmatrix} \times \begin{bmatrix} 0.4 & 0.4 & 0.2 \\ 0.1 & 0.4 & 0.5 \\ 0 & 0.6 & 0.4 \end{bmatrix}$$

$$= \begin{bmatrix} (0.08 + 0.02) & (0.2 + 0.2 + 0.3) & (0.06 + 0.15 + 0.12) \end{bmatrix}$$

$$= \begin{bmatrix} 0.13 & 0.46 & 0.41 \end{bmatrix}$$

Steady state probabilities (limiting probab.)

$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{bmatrix}$$

$$= \begin{bmatrix} 0.67 & 0.33 \\ 0.66 & 0.34 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.67 & 0.33 \\ 0.66 & 0.34 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{bmatrix}$$

$$= \begin{bmatrix} 0.667 & 0.333 \\ 0.666 & 0.334 \end{bmatrix}$$

$\frac{2}{3} \qquad \frac{1}{3}$

★ Converging the above

$$\pi = \begin{bmatrix} 2 & 1 \\ 3 & 3 \end{bmatrix}$$

$$P_i(n) \rightarrow \pi_i \text{ as } n \rightarrow \infty$$

$$\sum \pi_i = \pi_i \text{ and } \sum \pi_i = 1$$

If the TPM of Markov chain is

$$P = \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$
 Find its steady state matrix (π). Solve upto power 8 ($n=8$)

$$P = \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\pi = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.25 & 0.75 \\ 0.375 & 0.625 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} 0.375 & 0.625 \\ 0.3125 & 0.6875 \end{bmatrix}$$

$$P^5 = \begin{bmatrix} 0.3125 & 0.6875 \\ 0.3437 & 0.6562 \end{bmatrix}$$

$$P^6 = \begin{bmatrix} 0.3437 & 0.6562 \\ 0.3281 & 0.6718 \end{bmatrix}$$

$$P^7 = \begin{bmatrix} 0.3281 & 0.6718 \\ 0.3359 & 0.664 \end{bmatrix}$$

$$P^8 = \begin{bmatrix} 0.3359 & 0.664 \\ 0.332 & 0.6679 \end{bmatrix}$$

Poisson distribution Unit-5

Poisson process

$$P[X(t)] = \frac{e^{-\lambda t} \cdot (\lambda t)^n}{n!}; n=0,1,2,\dots$$

Mean and variance of poisson process

$$M = \lambda t \quad V = \lambda t$$

Find autocorrelation $[R_X(t_1, t_2)]$

$$R_X(t_1, t_2) = \lambda^2 t_1 t_2 + \lambda t_1 \quad \text{? solve in detail}$$

$$C_X(t_1, t_2) = \lambda t_1$$

If a bank receives an average 6 bad cheques per day. What are the probability that it will receive

- i) 4 bad cheques on any given day
- ii) 10 bad cheques on any 2 consecutive days

i) $\lambda = 6$; $\lambda t = 6$

$$t = 1$$

$$n = 4$$

$$P[X(1)=4] = \frac{e^{-6} \cdot (6)^4}{4!} = 0.134$$

ii) $\lambda = 6$; $\lambda t = 12$

$$t = 2$$

$$n = 10$$

$$P[X(2)=10] = \frac{e^{-12} \cdot (12)^{10}}{10!} = 0.105$$

$$\int_0^{\infty} t^{2-1} e^{-t} dt =$$

Customers arrive at a counter in accordance with poisson process with mean rate of 2 per minute.

Find the probability that the interval between 2 consecutive arrivals is

- i) more than 1 minute
- ii) between 1 and 2 minutes
- iii) 4 minutes or less

i) By the property of Poisson process, the time of interval T between two consecutive customers follows exponential distribution with $\lambda=2$, $t=1$

$$f(t) = \lambda e^{-\lambda t}$$

$$= 2e^{-2t}$$

$$P(X(t) > 1) = \int_1^{\infty} 2e^{-2t} dt$$

$$= 2 \left[\frac{e^{-2t}}{-2} \right]_1^{\infty}$$

$$= 2 \left[\frac{e^{-2t}}{-2} + \frac{e^{-2}}{2} \right]$$

$$= 0.1353$$

$$\approx 0.135$$

$$\text{ii } P(1 > t > 2) = \int_1^2 2e^{-2t} dt$$

$$= 2 \left[\frac{e^{-2t}}{-2} \right]_1^2$$

$$= 2 \left[\frac{e^{-4}}{-2} + \frac{e^{-2}}{2} \right]$$

$$= 0.117$$

$$\text{iii } P(t \leq 4) = \int_0^4 2e^{-2t} dt$$

$$= 2 \left[\frac{e^{-2t}}{-2} \right]_0^4$$

$$= 2 \left[\frac{e^{-8}}{-2} + \frac{e^{-0}}{2} \right]$$

$$= 0.9996$$

$$\approx 0.999$$

A radioactive substance emits particles at the rate of 6 particles per minute according to poisson process. The probability that the emitted particles will be recorded as 0.5. Find the probability that 10 particles will be recorded in 4 minutes time period.

$$\begin{aligned}
 P(N(t)=n) &= \frac{e^{-\lambda pt} \cdot (\lambda pt)^n}{n!} ; \lambda=6, p=0.5, t=10 \\
 &= \frac{e^{-(6 \times 4 \times 0.5)} \cdot (6 \times 4 \times 0.5)^{10}}{10!} \\
 &= 0.1048 \\
 &\approx 0.104
 \end{aligned}$$

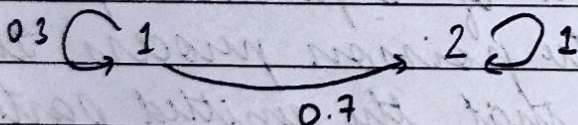
Continuous time Markov Chain Unit 5

TPM / TDiag $\rightarrow$ Create a TPM, Create a Diag

what are communicate states and why?

Aperiodic & periodic ??

$$P = \begin{array}{c|cc} & 1 & 2 \\ \hline 1 & 0.3 & 0.7 \\ 2 & 0 & 1 \end{array} \quad \begin{array}{l} \bullet i \rightarrow j \\ \bullet \text{self loop} = 1 \end{array}$$



1 $\rightarrow$ 2 accessible state

2 $\nrightarrow$ 1 not accessible

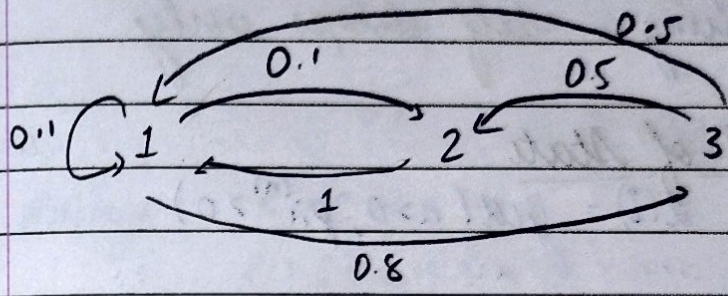
2 $\rightarrow$ 2 accessible (self loop)

1 $\rightarrow$ 1 not accessible

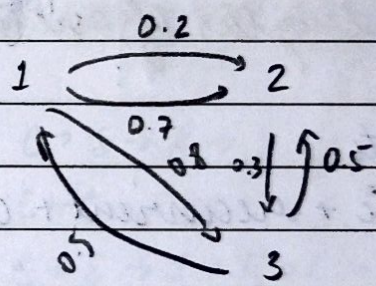
Communicate state

	1	2	3
1	0.1	0.1	0.8
2	1	0	0
3	0.5	0.5	0

$i \rightarrow j \text{ \& } j \rightarrow i$



Irreducible

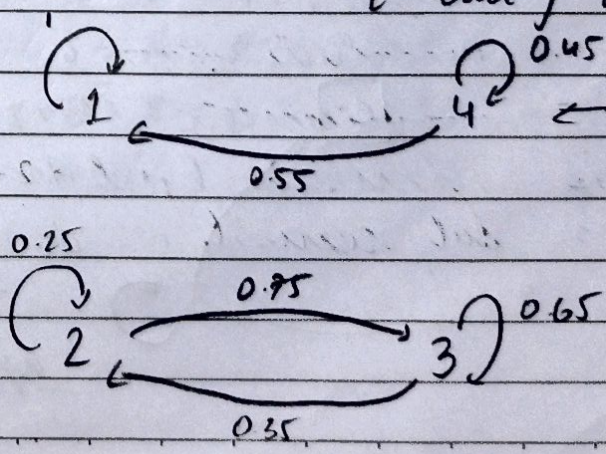


there is only
 one class
 [total should be 1]

all states should
 communicate

Transient

$i \text{ but } j \text{ where } j \neq i$



Transient
 from 4 $\rightarrow$ 1

but not
 coming
 back

Recurrent (transient)
like communicate but should come back

Last diagram, 2 & 3 are recurrent

absorbing self loops only.

Period of State

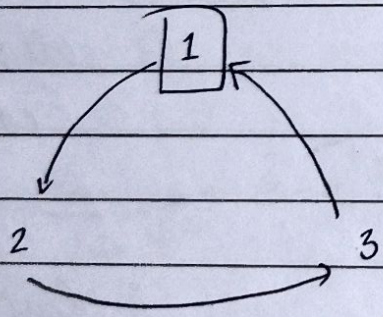
$$d(i) = \gcd(n > 0, p_{ii}^{(n)} > 0)$$

if $d(i) = 1$, aperiodic $d(i) = 0$,
~~else $d(i) = 0$~~ , periodic bridging state
 { cycle lengths of one member }

Ergodic state

$\Rightarrow$ aperiodic + recurrent + communicate

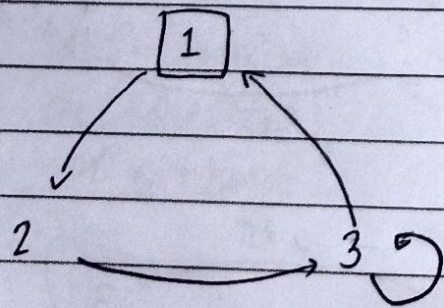
state 1



$1 \rightarrow 2 \rightarrow 3 \rightarrow 1$

$$d(1) = 3$$

periodic



$1 \rightarrow 2 \rightarrow 3 \rightarrow 1 = 3$

$1 \rightarrow 2 \rightarrow 3 \rightarrow 3 \rightarrow 1 = 4$

$$d(1) = \gcd(3, 4)$$

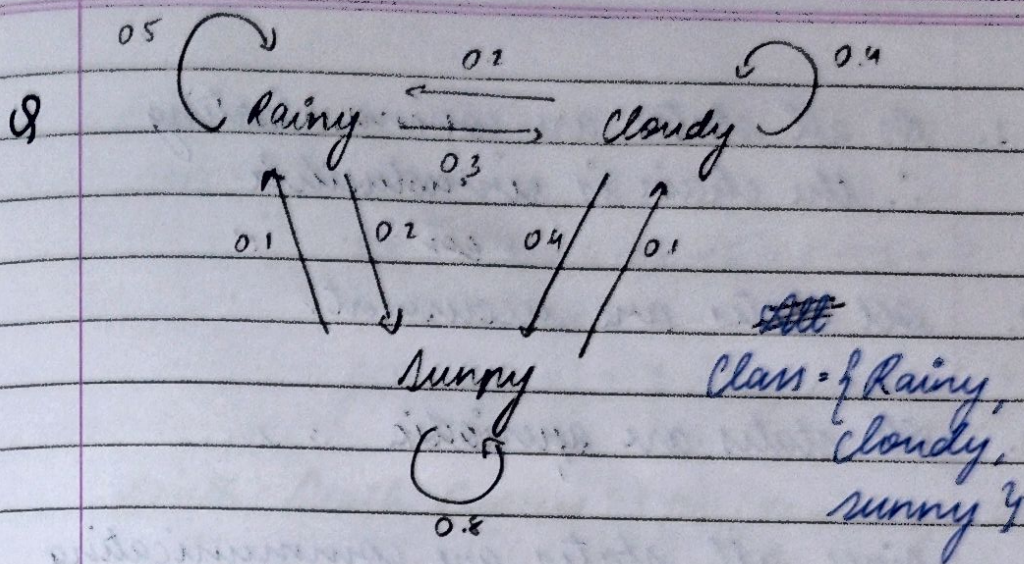
$$= 1$$

aperiodic

maximum 3 steps

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Rainy: $R \rightarrow C \rightarrow S \rightarrow R = 3$

$R \rightarrow C \rightarrow C \rightarrow S \rightarrow R = 4$

$R \rightarrow C \rightarrow S \rightarrow S \rightarrow R = 4$

$R \rightarrow R = 1$

~~C → S~~ $\gcd(3, 4, 4, 1) = 1$ aperiodic

cloudy: $C \rightarrow S \rightarrow R \rightarrow S = 3$

$C \rightarrow C = 1$

$C \rightarrow S \rightarrow S \rightarrow R \rightarrow C = 4$

$C \rightarrow S \rightarrow R \rightarrow R \rightarrow C = 4$

$\gcd(3, 4, 4, 1) = 1$

Sunny: $S \rightarrow R \rightarrow C \rightarrow S = 3$

$S \rightarrow S = 1$

$S \rightarrow R \rightarrow R \rightarrow C \rightarrow S = 4$

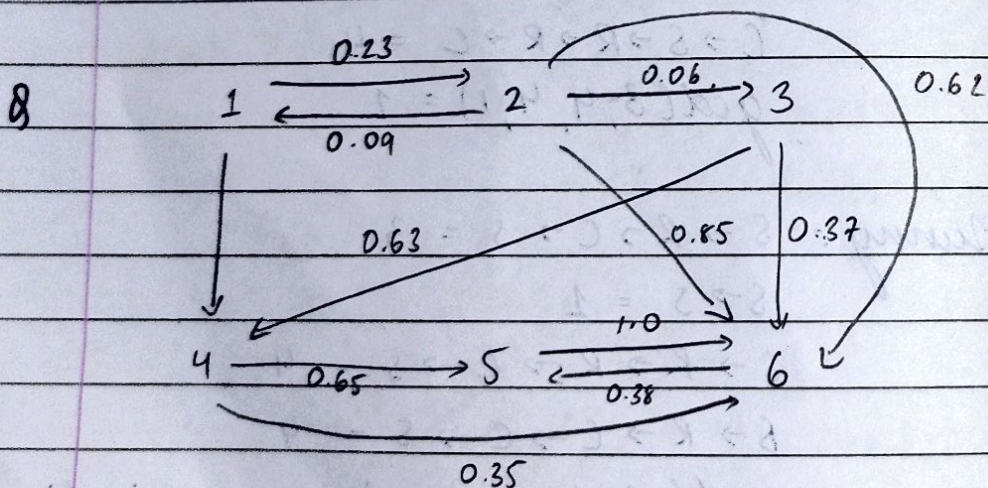
$S \rightarrow R \rightarrow C \rightarrow C \rightarrow S = 4$

$\gcd(3, 4, 4, 1) = 1$ aperiodic

1. As all states are communicating
 $\therefore$ the chain is irreducible
2. All states are recurrent
3. All states are aperiodic

Since all states are communicating, recurrent and aperiodic, this is an ergodic state.

	C	R	S
C	0.4	0.3	0.3
R	0.5	0.3	0.2
S	0.5	0.1	0.4



* Communicate state (make classes)
 Class 1: $\{1, 2, 3, 4, 5, 6\}$
 As all states are communicating
 $\therefore$ the chain is irreducible

Periodicity

state 1: $1 \rightarrow 2 \rightarrow 1 = 2$

$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 2 \rightarrow 1 = 7$

$1 \rightarrow 2 \rightarrow 6 \rightarrow 2 \rightarrow 1 = 4$

$\gcd(2, 7, 4) = 1$ {aperiodic} M2

Birth-Death Process (How to manage Queue)

- continuous time markov chain
- Queuing theory (virtual networking)

M1 $\rightarrow$ M/M/1 $\leftarrow$ one server N/FIFO $\Rightarrow$ Doctor

person on process

exponential distribution

finite no of people in queue

M2 $\rightarrow$

M/M/1

∞ / FIFO $\Rightarrow$ Airport

infinite people in the loop

1. λ = Mean arrival rate
2. μ = Mean service rate
3. L_s = Expected service length
4. L_q = Expected Queue length
5. W_s = Expected waiting time per customer in service
6. W_q = Expected waiting time per customer in queue
7. L_n = Expected length of non-empty queue (Average length)

Arrival rate = λ / hr γ mean
 Service rate = μ / hr γ mean
 Traffic intensity = ~~λ/μ~~ $\rho = \lambda/\mu$
 and $\rightarrow P_n = \rho^n (1-\rho)$

* Average number of customers in service
 $L_s = \lambda / \mu - \lambda$

* Average number of customers in queue
 $L_q = \lambda^2 / \mu(\mu - \lambda)$

* Average waiting time of customer in queue
 $W_q = \lambda / \mu(\mu - \lambda)$

* Average waiting time of customer in service
 $W_s = 1 / \mu - \lambda$

* Probability of queue size exceeds 'N'
 $P(\text{queue size} \geq N) = \rho^N$

* Expected length of non-empty queue
 $L_n \Rightarrow \mu / \mu - \lambda = 1 / 1 - \rho$

* Probability that no customer at the counter
 is zero $P_0 = 1 - \rho = 1 - \lambda / \mu = \mu - \lambda / \mu$
 $P_0(1 - \rho)$

* Probability of waiting time in queue $\geq N$
 $P(\text{waiting time} \geq N) = \int_N^{\infty} \frac{\lambda(1-\lambda)}{\mu} e^{-(\mu-\lambda)t} dt$

- * Probability that an arrival will have to wait for more than 'N' minutes

$$P(\text{waiting time} + \text{service time} \geq N) = P^N$$

$$P^N = \int_N^{\infty} (M - \lambda) e^{-(M - \lambda)t} dt$$

Little's formula

1. $L_q = L_s - \frac{\lambda}{M}$

2. $W_q = W_s - \frac{1}{M}$

3. $L_s = \lambda W_s$

4. $L_q = \lambda W_q$

- Q. Customers arrive at a loan office with 1 ticket window according to poisson's input process with mean rate of 30/hr. The time required to serve a customer has an exponential distribution with mean 90 seconds. Find the average

- i) Length of service
- ii) Queue length
- iii) Waiting time in queue
- iv) Time spent by a customer in the system

1
n

0.025

1

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$$n = 40$$

$\lambda = 30$ customers per hour

$\mu = 1/40 \text{ sec} = 1.5 \times 60 \times 60 \text{ services per hour}$

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{30}{40 - 30} = 3$$

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{30^2}{40(40 - 30)} = 2.25$$

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{30}{40(40 - 30)} = \frac{3}{40} \text{ hr} = 4.5 \text{ min}$$

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{40 - 30} = \frac{1}{10} \text{ hr} = 6 \text{ min}$$

i) The time spent by a repair man on his jobs has an exponential distribution with mean as 30 minutes. If he repairs sets in the order in which they come in and if the arrival of sets is approximately poisson with an average rate of 10 per 8-hr day (10/8 hr day). what is the repair man's expected idle time each day.

ii) How many jobs are ahead of the average set just brought in?

$$\frac{1}{n} \quad \bigg| \quad \frac{0.5}{1}$$

$$\lambda = 10/8 \text{ hr day} = \textcircled{10} - \frac{10 \times 8}{8} \text{ (because 8 hr day)}$$

because M is in 8 hour day we don't consider 18 in λ

$$\rightarrow \mu = 30 \text{ minutes} = 0.5 \text{ hrs} = 2 \times 8 = \textcircled{16} \text{ per hour}$$

$$\rho = \lambda/\mu = 10/16 = 5/8 = 0.625$$

ii) $P_0 = 1 - \rho = 1 - \lambda/\mu = 1 - 10/16 = 6/16$
 $= 3/8 \text{ part of a day}$

$$= 3$$

iii) $L_s = \frac{\lambda}{\mu - \lambda} = \frac{10}{16 - 10} = \frac{5}{3} \Rightarrow 1 \frac{2}{3} \text{ set}$

Q Arrivals at a telephone booth are considered to be poisson with an average time of 10 minutes between 1 arrival and the next. The length of a phone call assumed to be distributed exponentially with mean 3 minutes. Find the following:

i) what is the probability that the person arriving at the booth will have to wait

ii) what is the average length of the queue that forms from time-to-time.

iii) The telephone department will install a second booth when convinced that an arrival would expect to have to wait atleast 3 minutes for the phone. By how much must the flow of arrivals

be increased in order to justify a second booth.

$$\lambda = 10 \text{ minutes} = \frac{10}{60} = 0.6 = \frac{1}{0.6} = 6$$

$$\mu = 3 \text{ minutes} = \frac{3}{60} = 0.20 = \frac{1}{0.20} = 20$$

$$\begin{aligned} \text{ii) } P(\text{queue size} \geq 1) &= \rho^N = \rho' \\ &= \left(\frac{\lambda}{\mu} \right)' = \frac{6}{20} = 0.3 \end{aligned}$$

$$\text{ii) } L_n \Rightarrow \frac{\mu}{\mu - \lambda} = \frac{20}{20 - 6} = \frac{10}{7} = 1.428$$

$$\text{iii) } W_q = 3 \text{ minutes} = \frac{3}{60} = 0.20 \text{ hour}$$

$$3 = \frac{\lambda'}{\mu(\mu - \lambda')} = \frac{\lambda'}{20(20 - \lambda')}$$

$$3 = \frac{\lambda'}{\mu^2 - \mu\lambda'} \quad \frac{\lambda'}{3} = 20^2 - 20\lambda'$$

$$\lambda' = 1200 - 60\lambda'$$

$$\lambda'(1 + 60) = 1200$$

$$\lambda' = \frac{1200}{61} = 19.672$$

As there are 13 more arrivals exceeding the queue, hence we should introduce booth 2.